MILITARY SPECIFICATION

TEST SET, ELECTRONIC CIRCUIT PLUG-IN UNIT

AN/USM-146(*)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The test equipment covered by this specification shall be for shop use and shall provide facilities specified herein for testing the performance of eleven individual Radio Set AN/ARC-80 Modules of the AN/ASQ-52.

Items	Type Designation	Appl. Para.
Test Set, Electronic Circuit Plug-in Unit	TS-(*)/USM-146	3.4.1
Case, Test Set	CY-(*)/USM-146	3.4.2
Cable, Power	W-1 of AN/USM-146(*)	3.4.3

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and application of
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment

FSC 6625

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MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedure: Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Specification for
MIL-M-7793	Meter, Time Totalizing
MIL-D-23647	Data Communication System AN/ASQ-52,

STANDARDS, Military

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-242	Electronic Equipment Parts
MIL-STD-781	Reliability Tests Exponential Distribution
MS-90287	Lights, Indicator, Miniature Styles 62 and 64
MS-91528	Knobs, Control, Round, Plastic

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of the specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

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3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied as outlined in Specification MIL-E-4682.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction and workmanship except as otherwise specified herein.

3.2.1 Total Weight - The total weight of the equipment, including cables, and the transmit case shall be a minimum consistent with good design and shall not exceed 70 pounds.

3.2.2 Reliability

3.2.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 100 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.2.2 Operating Life - The equipment shall have a minimum total operating life of 2,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

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3.2.2.3 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have a minimum of 100 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.2.2.4 Time Totalizing Meter - The equipment shall contain a time totalizing meter in accordance with Specification MIL-M-7793 and MS-24534.

3.2.2.5 Modules, Maintenance - The electronic portions of the equipment shall be ddvided into maintenance modules (see Specification MIL-S-23603). The maintenance modules shall normally be considered repairable.

3.2.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.2.3.1 Cable Assemblies, Power, Electrical - Primary power cable assembly for this equipment shall conform with Specification MIL-T-21200.

3.2.4 Interchangeability - The equipment shall meet the interchangeability requirement as defined in Specification MIL-T-21200.

3.2.5 Interference Control - The generation of conducted radio interference by the equipment and the vulnerability of the equipment to conducted radio interference shall be controlled within the limits of Specification MIL-I-6181. The requirements of Specification MIL-I-6181 concerning radiated interference shall not apply except for radiation susceptability.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.6.1 Maintenance Positions - The set shall be capable of standing without additional support on at least four sides.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{\circ} \pm 10^{\circ}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input power voltage	115 ± 1.0 VAC, 380 to 420 cps

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3.2.9 Service Conditions - The test equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 3 equipment, except as modified herein.

3.2.9.1 Temperature - The equipment shall operate at any temperature between -20°C and +40°C at any altitude from sea level to 10,000 feet. Non-operating equipment shall not be damaged by exposure to temperatures ranging from -62°C to +85°C at altitudes from sea level to 50,000 feet.

3.2.10 Warm-Up Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 2 minutes under standard conditions and 5 minutes at extreme service conditions, except than when the r-f oscillator is used, warm-up time shall be 20 minutes.

3.2.11 Primary Input Power Requirements - The test equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in Specification MIL-T-21200 except that operation from a 60 cycle power source shall not be required. The power required shall be of the following type and shall not exceed specified amount.

AC Power, Three Phase, 4 wire, 115V, 400 ± 20 cps, 170 VA maximum

3.2.11.1 Voltage Protection - The equipment shall not be damaged when exposed to transient as defined in Figure II of MIL-STD-704, and shall automatically resume normal operation when voltage returns within the limits of normal operation.

3.2.12 Standby Provisions - Standby operation of the equipment shall not be required.

3.2.14 Fungus Prevention - Prevention of deterioration due to fungus shall be accomplished by selection of parts and materials which are non-nutrient for fungus, or the parts and materials shall be treated prior to their use in the equipment so that overall treatment of the equipment after assembly is not required.

3.2.17 Exception for Subassemblies - The requirements of this specification shall not apply to plug-in modules of Data Communication System AN/ASQ-52 used in Test Set, Electronic Circuit Plug-In Unit, AN/USM-146(*). These modules shall be in accordance with Specification MIL-D-23647 for Data Communication System AN/ASQ-52 and shall be identical to production AN/ASQ-52 modules except that the Power Supply Module shall be modified to provide for external control of the +18 VDC supply voltage. The test set, including these units, shall meet all performance and environmental requirements specified herein.

3.2.18 Legends - Characters in all front panel legends shall be engraved and filled with white enamel.

3.2.19 Knobs - Knobs shall be in accordance with Standard MS-91528.

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3.2.20 Indicator Lights - Indicator lights shall be in accordance with Standard MS-90287.

3.3 Construction - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

3.3.1 Performance - Test Set, Electronic Circuit Plug-In Unit AN/USM-146(*), when used in conjunction with Test Bench, Data Communication System AN/USM-145 and the general purpose test equipment of 3.5, shall provide facilities to enable the operator to isolate a malfunction in any one of eleven individual modules of Radio Set AN/ARC-80 to the nonrepairable subassembly level and to align and test a repaired module. The alignment and calibration capabilities provided by the test set shall be of sufficient accuracy to insure that, at the time a repaired module is returned to service, additional calibration or alignment of the module shall not be required to maintain AN/ASQ-52 system performance within the limits of Specification MIL-D-23647.

3.4 Detail Requirements

3.4.1 Electronic Circuit Plug-In Unit TS-(*)/USM-146- The test set shall meet the following requirements:

3.4.1.1 Function - The test set when used with Test Bench, Data Communication System AN/USM-145 and the general purpose test equipment of 3.5 shall provide facilities for test and alignment of the following AN/ASQ-52 system modules:

(1) Balanced Modulator, Module 7A8 of Transmitter, Radio, T-899/ARC-80

(2) R-F Tuner, Module 2A1 of Receiver, Radio, R-1153/-ARC-80 and Module 7A1 of Transmitter, Radio, T-899/-ARC-80

(3) R-F Oscillator, Module 7A4 of Transmitter, Radio T-899/ARC-80

(4) Frequency Divider, Module 2A3 of Receiver, Radio R-1153/ARC-80 and Moduel 7A3 of Transmitter, Radio, T-899/ARC-80

(5) Frequency Multiplier, Module 2A4 of Receiver, Radio, R-1153/ARC-80

(6) Receiver Gain Control, Module 2A7 of Receiver, Radio, R-1153/ARC-80

(7) Transmitter Gain Control, Module 7A7 of Transmitter, Radio, T-899/ARC-80

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(8) Frequency Stabilizer (KC), Module 2A5 of Receiver, Radio, R-1153/ARC-80 and Module 7A5 of Transmitter, Radio, T-899/ARC-80

(9) Frequency Stabilizer (MC), Module 2A2 of Receiver, Radio, R-1153/ARC-80 and Module 7A2 of Transmitter, Radio, T-899/ARC-80

(10) Amplifier-Mixer, LSB, Module 2A6 of Receiver, Radio, R-1153/ARC-80

(11) Amplifier-Mixer, USB, Module 2A8 of Receiver, Radio, R-1153/ARC-80

The test set shall provide a physical mount for the module under test, power and controlled signal inputs to the module, and loads and test points on the module outputs. Module input signals shall be derived from internal generating circuits, from signals supplied by the AN/USM-145 with its AN/ASQ-52 system installed, or from general purpose test equipment.

3.4.1.2 Form Factor - The Test Set TS(*)/USM-146 shall be packaged in a portable instrument case in accordance with Specification MIL-T-21200. Controls and indicators shall be located on the front panel, which shall be inclined to provide an optimum viewing angle when the set is used on a bench. Recessed carrying handles shall be provided on the sides of case. The degree of enclosure of the instrument case shall be "protected" in accordance with MIL-STD-108. The maximum case dimensions shall be 23-3/4 inches wide by 17-3/8 inches deep by 8-1/4 inches high.

3.4.1.3 Weight - The weight of the test set, exclusive of the transit case and cable, shall not exceed 43 pounds.

3.4.1.4 Contents of Unit - The test set shall contain the following assemblies:

MODULES

(1) AC Power Supply - The AC Power Supply Module shall supply the following voltages for use in the test set and the module under test:

6.3 volts AC ± 13 percent
26 volts AC ± 13 percent
26 volts DC ± 13 percent
18 volts DC ± 1 percent
130 volts DC ± 3 percent
250 volts DC ± 10 percent

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Input to the AC Power Supply Module shall be 115 volts AC, 400 cps, 3 phase. This module shall be similar to the AC Power Supply used in Radio Set AN/ARC-80 except that the 18 volt DC level shall be controlled from an external source and the 250 volt DC supply shall have an increased power handling capability.

(2) Variable Frequency Oscillator - The Variable Frequency Oscillator module shall provide a range of frequencies between 2501 kilocycle and 3500 kilocycles to the (KC) Frequency Stabilizer module under test. Control of the Variable Frequency Oscillator output frequency shall be provided by a control on the front panel of the test set.

(3) Signal Generator - The Signal Generator module shall provide a variable frequency 500 kilocycle signal used in testing the following modules:

USB Amplifier - Mixer
LSB Amplifier - Mixer
Receiver Gain Control
Transmitter Gain Control

In addition, the Signal Generator module shall provide a variable frequency 100 kilocycle signal used in testing the Frequency Divider module. Control of the amplitude of the 100 KC and 500 KC signal from the Signal Generator module shall be provided by a control on the front panel of the test set.

(4) Audio-Frequency/Radio-Frequency Amplifier - The A-F/R-F Amplifier shall be capable of amplifying either 500 KC from the Balanced Modulator and the Transmitter Gain Control modules under test or audio from the USB Amplifier-Mixer and LSB Amplifier-Mixer modules under test. The A-F/R-F Amplifier shall provide a minimum of 28 db of gain at 500 KC. In addition, it shall provide isolation between the output of the module under test and the external test equipment. Intermodulation distortion shall be a minimum of 52 db below either tone of a two tone audio test signal.

(5) Frequency Divider - The Frequency Divider module shall provide a 10 kilocycle pulse and a 1 kilocycle spectrum to the (KC) Frequency Stabilizer module under test. This module shall be identical to the Frequency Divider module used in Radio Set AN/ARC-80.

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(6) (MC) Frequency Stabilizer - The (MC) Frequency Stabilizer module shall provide the frequency stabilization circuitry for the h-f (8-16 mc) oscillator and the 17.5 mc oscillator of the R-F Turner module under test. This module shall be identical to the (MC) Frequency Stabilizer module used in Radio Set AN/ARC-80.

(7) Permeability Tuned Oscillator - The Permeability Tuned Oscillator module shall provide an 1850 kilocycle injection signal to the Signal Generator module of the test set. Control of the output frequency of the Permeability Tuned Oscillator shall be provided by a control on the front panel of the test set.

CIRCUITS

(8) R-F Oscillator Module - Provision shall be made within the test set to mount an R-F Oscillator module from Radio Set AN/ARC-80 to be used as a source of stable 100 KC and 500 KC in the event these signals cannot be obtained from Test Bench, Data Communication System AN/USM-145 or other external sources. The R-F Oscillator module is not supplied as a part of the Test Set, Electronic Circuit Plug-In Unit, AN/USM-146(*).

(9) Module Selector Switch - A Module Selector Switch shall be provided to connect various signals and voltages to the individual Radio Set AN/ARC-80 modules under test.

(10) Frequency Control Switches - Frequency control switches shall be provided to exercise the frequency selecting and stabilizing circuits of the individual Radio Set AN/ARC-80 modules under test.

3.4.1.5 Controls - The following controls shall be provided on Test Set, Electronic Circuit Plug-In Unit AN/USM-146(*).

(1) Module Test Selector Switch - A rotary switch shall be provided to select the following module test conditions:

OSCILLATOR 1
OSCILLATOR 2
MC STABILIZER
R-F TUNER 1
R-F TUNER 2
R-F TUNER 3
KC STABILIZER 1
KC STABILIZER 2
FREQ DIVIDER

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FREQ MULTIPLIER
USB AMPLIFIER
LSB AMPLIFIER
BAL MODULATOR 1
BAL MODULATOR 2
TRANSMITTER GAIN CONTROL 1
TRANSMITTER GAIN CONTROL 2
TRANSMITTER GAIN CONTROL 3
RECEIVER GAIN CONTROL 1
RECEIVER GAIN CONTROL 2
RECEIVER GAIN CONTROL 3

(2) FREQUENCY Selector Switches - Four switches shall be provided to exercise the frequency selection and stabilization circuits of the modules under test.

(3) KEY Switch - The KEY switch shall provide on-off keying control to the R-F Tuner module, Transmitter Gain Control module and Balanced Modulator modules under test.

(4) OSC CURRENT Switch - An OSC CURRENT switch shall be provided to open the 28 volt d-c line to the R-F Oscillator module under test to allow for external connection of an ammeter between the 28 volt d-c source and the R-F Oscillator.

(5) T/R Switch - The T/R switch shall provide transmit-receive control information to the R-F Tuner module under test.

(6) UNLOCK Switch - The UNLOCK button, when depressed, shall remove the (KC) Frequency Stabilizer module control of the frequency of the integral Variable Frequency Oscillator module.

(7) ATTENUATOR Switch - The ATTENUATOR switch permits insertion of a fixed 80 db attenuator into the input circuit to the USB Amplifier-Mixer or LSB Amplifier-Mixer modules under test. It also permits insertion of a fixed 6 db attenuator into the input circuit to the Receiver Gain Control module under test.

(8) MOTOR Switch - The MOTOR switch shall provide forward-reverse information to the servo motor of the Transmitter Gain Control module under test.

(9) 2 TONE Switch - A 2 TONE switch shall be provided which will allow for the selection of an externally supplied 500 kilocycle 2 tone signal when testing the USB or LSB Amplifier-Mixer modules.

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(10) POWER Switch - A POWER switch shall be provided which will control the primary power to the test set.

(11) AGC Control - A potentiometer shall be provided which will control the AGC voltage for the R-F Tuner module, USB Amplifier-Mixer module, LSB Amplifier-Mixer module, and the Transmitter Gain Control module under test.

(12) ALC Control - A potentiometer shall be provided which will control the ALC voltage supplied to the R-F Tuner module or the Transmitter Gain Control module under test.

(13) DC SET Control - The DC SET potentiometer shall provide for the variation and adjustment of the 18 volt d-c power supply voltage.

(14) 500 KC AMPLITUDE Control - A potentiometer shall be provided to adjust the amplitude of the stable 500 kilocycle signal.

(15) VFO BIAS Control - A potentiometer shall be provided to control the amplitude of the VFO Bias supplied to the R-F Tuner module under test.

(16) R-F TUNER GAIN Control - A potentiometer shall be provided to control the gain of the R-F Tuner module under test.

(17) OSCILLATOR AMPLITUDE Control - A potentiometer shall be provided to adjust the amplitude of the output signal from the internal Signal Generator module.

(18) VFO Control - The VFO control varies the output frequency of the internal Variable Frequency Oscillator module.

(19) FREQUENCY DEVIATION KC Control - This control provides for a ± 10 kilocycle variation of the output frequency of the internal Permiability Tuned Oscillator module.

3.4.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	MS-3102R-14S-5P	Primary Power Input
J2	DBMF-13W3S or equal	Supplies signals to (MC) Frequency Stabilizer module under test

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Reference Designation	Receptacle Type	Function
J3	DEF-9S-C7-A101 or equal	Supplies signals to R-F Oscillator module under test
J4	DAMF-3W3S or equal	Supplies signals to R-F Oscillator module under test
J5	DAMF-3W3S or equal	Supplies signals to R-F Oscillator module under test
J6	DEF-9S-C7-A101 or equal	Supplies signals to Frequency Multiplier module under test
J7	DAMF-3W3S or equal	Supplies signals to Frequency Multiplier module under test
J8	DAMF-3W3S or equal	Supplies signals to Frequency Multiplier module under test
J9	DCF-37S-C7-A101 or equal	Supplies signals to R-F Tuner module under test
J10	DBF-25S-C7-A101 or equal	Supplies signals to R-F Tuner module under test
J11	DAMF-3W3S or equal	Supplies signals to R-F Tuner module under test
J12	DAMF-3W3S or equal	Supplies signals to R-F Tuner module under test
J13	DAMF-3W3S or equal	Supplies signals to R-F Tuner module under test
J14	DBMF-13W3S or equal	Supplies signals to (KC) Frequency Stabilizer module under test
J15	DBMF-13W3S or equal	Supplies signals to Frequency Divider module under test
J16	DBMF-13W3S or equal	Supplies signals to LSB Amplifier-Mixer module under test
J17	DBMF-17W2S or equal	Supplies signals to RCVR Gain Control module under test

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Reference Designation	Receptacle Type	Function
J18	DAF-15S-C7-A101 or equal	Supplies signals to Balanced Modulator module under test
J19	Special Coaxial	Supplies signals to Balanced Modulator module under test
J20	Special Coaxial	Supplies signals to Balanced Modulator module under test
J21	Special Coaxial	Supplies signals to Balanced Modulator module under test
J22	DAMF-3W3S or equal	Supplies signals to Transmitter Gain Control module under test
J23	DBF-25S-C7-A101 or equal	Supplies signals to Transmitter Gain Control module under test
J32	DBMF-13W3S or equal	Supplies signals to USB Amplifier-Mixer module under test
J38	UG-909B/U	17.5 MC Input
J39	UG-909B/U	8-16 MC Input
J40	UG-909B/U	Receiver Output
J41	UG-909B/U	Transmitter Output
J42	UG-909B/U	Receiver Input
J43	UG-909B/U	Transmitter Output
J44	UG-909B/U	500 KC Input
J45	UG-909B/U	100 KC Input
J46	UG-909B/U	Input
J47	UG-909B/U	Output
J48	UG-909B/U	LSB Audio Input
J49	UG-909B/U	USB Audio Input

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Reference Designation	Receptacle Type	Function
J50	UG-909B/U	500 KC Two Tone Input
J51	UG-909B/U	A-F/R-F Signal Output
TP1	MS-16108-6	10V DC
TP2	MS-16108-7	Oscillator Current
TP3	MS-16108-5	Test Signal
TP4	MS-16108-4	Input
TP5	MS-16108-7	Output
TP6	MS-16108-1	500 KC
TP7	MS-16108-5	26 VAC
TP8	MS-16108-6	+130 VDC
TP9	MS-16108-8	+28 VDC
TP10	MS-16108-2	+18 VDC
TP11	MS-16108-2	+250 VDC
TP12	MS-16108-3	AGC
TP13	MS-16108-4	ALC

3.4.1.6.1 Location of Electrical Connections - Electrical connections for introduction of test signals and for connection of test equipment shall be located along the lower edge of the front panel. Test points shall be located along the right edge of the top surface. Receptacles for modules under test shall be located on the top surface of the unit.

3.4.2 Case, Test Set CY-(*)/USM-146 - The test equipment shall be provided with a transit case which shall conform to Specification MIL-T-21200.

3.4.2.1 Function - The case shall provide a convenient means for transporting and storing the test set.

3.4.2.2 Form Factor - The maximum case dimensions shall be 27-3/4 inches wide, 21-11/16 inches deep and 11-1/8 inches high.

3.4.2.3 Weight - The weight of the case shall not exceed 26 pounds.

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3.4.2.4 Stowage - The case shall provide stowage space for Primary Power Cable W1, and the equipment handbook.

3.4.3 Cable, Primary Power, W-1 of Test Set, Electronic Circuit Plug-In Unit AN/USM-146(*) - The test set shall be provided with a primary power cable as outlined below.

3.4.3.1 Form Factor - Cable W-1 shall have a minimum length of 60 inches including connectors.

3.4.3.2 Weight - Cable W-1 shall have a maximum weight of 0.4 pound.

3.4.3.3 Connectors - Cable W-1 shall be provided with the following electrical connectors:

Reference Designation	Receptacle Type	Function
P1	MS-3106A-14S-5S	115V to Test Set
P2	PT06A-14-5P	Mates to 115V 400 cps, 3 phase connector on Test Bench, AN/USM-145

3.5 Associated Equipment - The test set shall operate with the following associated equipment which shall not be supplied as part of the test set:

(1) Test Bench, Data Communication System AN/USM-145 or equivalent

(2) AC VTVM, ME-30A/U or equivalent

(3) Hewlett Packard Model 411A or equivalent

(4) DC Differential Voltmeter, accuracy 0.05% of input, Fluke Model 801 or equivalent

(5) Counter, AN/USM-26 or equivalent

(6) Audio Oscillator, TS-382 D/U, Hewlett-Packard 200 CD or equivalent

(7) Data Communication System, AN/ASQ-52 or equivalent

(8) Oscilloscope, USM-105 or equivalent

(9) Distortion Analyzer, Collins 476D-1 or equivalent

(10) AC/DC VTVM TS-505D/U or equivalent

(11) AC/DC Multimeter AN/PSM-4B, or equivalent

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3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6 The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction test shall include environmental tests in accordance with the procedures of Specification MIL-T-21200 and interference tests in accordance with Specification MIL-I-6181.

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4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper correction action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

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4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of

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satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plan for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hours period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
1 for each additional 200 or fraction	

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain enviromental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. A Qualification Phase and Sampling Phase Tests shall be conducted.

*Note: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

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4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase Test Level E except -54 to + 55°C shall be 0 to + 40°C shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase Test Level E modified shall be used. The Accept-Reject criteria for Test Plan VI shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime other equipments constructed shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment quality control etc, so that the entire contract quantity will show an accept decision.

4.4.3.3 Test Details - The test details such as the length of the length of the Test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test etc. shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

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4.4.4.2 Scope of Tests - Special Tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure (s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)

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4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit referency symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

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4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life test shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity for the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-R-18303 shall be sued as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the orginal shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - It is intended that this equipment be for shop use and forvide facilities specified here in for testing the performance of eleven (11) individual radio set AN/ARC-80 Modules.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115 ± 1. 0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

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6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically, sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therin. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*) when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 2.7. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.
(2) Selection of Applicable levels of packaging and packing (See 5.1)
(3) Data Requirements (See 3.6)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-81436(AS) TEST SET, ELECTRONIC CIRCUIT PLUG-IN UNIT AN/USM-146(*)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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